the way to trust KCL

TEST REPORT

1. No : CT12-51652
2. Client
   - Name : WOOJIN INDUSTRIAL SYSTEMS
   - Address : 613-6, Bangchuk-ri, Sari-myeon, Gosan-gun, Chungbuk, Korea
   - Date of Receipt : Oct. 31, 2012
   - Date of Issued : Nov. 12, 2012
3. Use of Report : Submission
4. Test Sample : ① Communication Control Panel(CCP), ② Driver PA Operation Box (DPO), ③ Passenger Information Display 1(PID-1), ④ Exterior Speaker(ESP), ⑤ Gooseneck Microphone(GNM), ⑥ Transformer(TRF), ⑦ANS Microphone(ANSM) ⑧ Interior Speaker (ISP)
5. Test Results

Refer to the attached sheets

Affirmation	Tested by Name : Lee Dong Kun D.K. Lee	Technical Manager Name : Jae Sang Kim

Our report apply only to the standards or procedures identified and to the sample(s) tested unless otherwise specified.
The test results are not indicative of representative of the qualities of the lot from which the sample was taken or of apparently identical or similar products.

Korea Conformity Laboratories President Jae Bin Song Jae Bin Song

The above test certificate is the accredited test results by Korea Laboratory Accreditation Scheme, which signed the ILAC-MRA.

Address : 153-803 459-28, Gasan-Dong, Geumcheon-Gu, Seoul, Korea 82-2-2102-2500
Result Inquiry : 82-2-2102-2776

TEST REPORT

[TEST RESULTS]

1. Vibration Test - Longlife Test

Test Conditions						Test Results		
Direction (Axis)	Frequency Range (Hz)	PSD Level $((m/s^2)^2/Hz)$	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	①	②	③
X	5	0.452	-	3.90	5.0	No Defects	No Defects	No Defects
	20	0.452	-					
	150	-	-6					
Y	5	0.250	-	2.90	5.0	No Defects	No Defects	No Defects
	20	0.250	-					
	150	-	-6					
Z	5	1.034	-	5.90	5.0	No Defects	No Defects	No Defects
	20	1.034	-					
	150	-	-6					

Test Conditions						Test Results		
Direction (Axis)	Frequency Range (Hz)	PSD Level $((m/s^2)^2/Hz)$	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	④	⑤	⑥
X	5	0.452	-	3.90	5.0	No Defects	No Defects	No Defects
	20	0.452	-					
	150	-	-6					
Y	5	0.250	-	2.90	5.0	No Defects	No Defects	No Defects
	20	0.250	-					
	150	-	-6					
Z	5	1.034	-	5.90	5.0	No Defects	No Defects	No Defects
	20	1.034	-					
	150	-	-6					

Test Conditions						Test Results	
Direction (Axis)	Frequency Range (Hz)	PSD Level ($(m/s^2)^2$/Hz)	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	⑦	⑧
X	5	0.452	-	3.90	5.0	No Defects	No Defects
	20	0.452	-				
	150	-	-6				
Y	5	0.250	-	2.90	5.0	No Defects	No Defects
	20	0.250	-				
	150	-	-6				
Z	5	1.034	-	5.90	5.0	No Defects	No Defects
	20	1.034	-				
	150	-	-6				

2. Shock Test

Test Conditions						Test Results		
Direction (Axis)		Acceleration (m/s^2)	Duration Time (ms)	Wave Form	Repetitive Times	①	②	③
X	+	50	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects
Y	+	30	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects
Z	+	30	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects

TEST REPORT

Test Conditions						Test Results		
Direction (Axis)		Acceleration (m/s²)	Duration Time (ms)	Wave Form	Repetitive Times	④	⑤	⑥
X	+	50	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects
Y	+	30	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects
Z	+	30	30	Half Sine	3	No Defects	No Defects	No Defects
	-				3	No Defects	No Defects	No Defects

Test Conditions						Test Results	
Direction (Axis)		Acceleration (m/s²)	Duration Time (ms)	Wave Form	Repetitive Times	⑦	⑧
X	+	50	30	Half Sine	3	No Defects	No Defects
	-				3	No Defects	No Defects
Y	+	30	30	Half Sine	3	No Defects	No Defects
	-				3	No Defects	No Defects
Z	+	30	30	Half Sine	3	No Defects	No Defects
	-				3	No Defects	No Defects

3. Vibration Test - Functional Test

Test Conditions						Test Results		
Direction (Axis)	Frequency Range (Hz)	PSD Level $((m/s^2)^2/Hz)$	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	①	②	③
X	5	0.0073	-	0.500	0.5	No Defects	No Defects	No Defects
	20	0.0073	-					
	150	-	-6					
Y	5	0.0041	-	0.370	0.5	No Defects	No Defects	No Defects
	20	0.0041	-					
	150	-	-6					
Z	5	0.0166	-	0.750	0.5	No Defects	No Defects	No Defects
	20	0.0166	-					
	150	-	-6					

Test Conditions						Test Results		
Direction (Axis)	Frequency Range (Hz)	PSD Level $((m/s^2)^2/Hz)$	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	④	⑤	⑥
X	5	0.0073	-	0.500	0.5	No Defects	No Defects	No Defects
	20	0.0073	-					
	150	-	-6					
Y	5	0.0041	-	0.370	0.5	No Defects	No Defects	No Defects
	20	0.0041	-					
	150	-	-6					
Z	5	0.0166	-	0.750	0.5	No Defects	No Defects	No Defects
	20	0.0166	-					
	150	-	-6					

TEST REPORT

Test Conditions						Test Results	
Direction (Axis)	Frequency Range (Hz)	PSD Level ($(m/s^2)^2/Hz$)	Slope (dB/octave)	Acceleration (RMS Value m/s^2)	Test Time (h)	⑦	⑧
X	5	0.0073	-	0.500	0.5	No Defects	No Defects
	20	0.0073	-				
	150	-	-6				
Y	5	0.0041	-	0.370	0.5	No Defects	No Defects
	20	0.0041	-				
	150	-	-6				
Z	5	0.0166	-	0.750	0.5	No Defects	No Defects
	20	0.0166	-				
	150	-	-6				

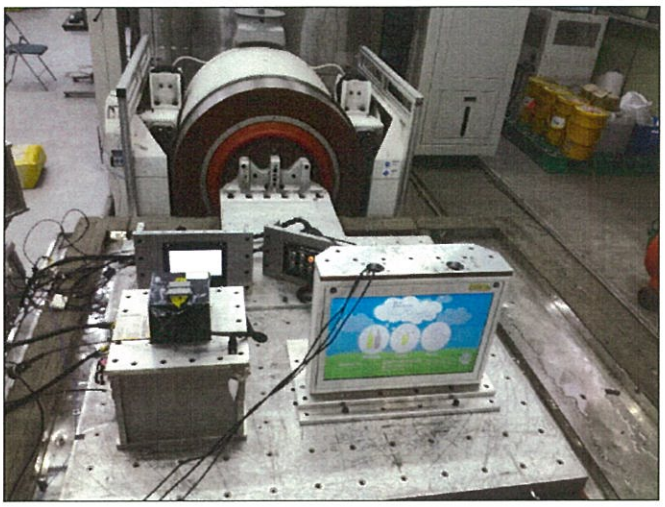
▶ Test Method : KS C IEC 61373 : 2002 Class 1-A

After the sample was made to be fastened tightly on the table of the vibration tester and to be taken tests against 3-Axis according to the above test conditions, to check whether major defects - like scratch, deformation, distortion of the sample - are found with a visual.
(But, refer to the returned samples about other physical characteristic changes.)

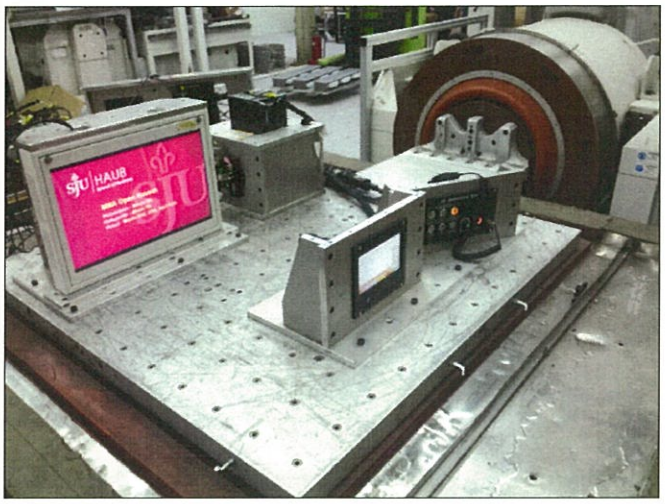
cf. Vibrator :

- Manufacturer : EMIC CO. (Japan), - Model : F-26000BDH/LA26AW
- Manufacturer : IMV CO. (Japan), - Model : CV-700, J255/SA7M

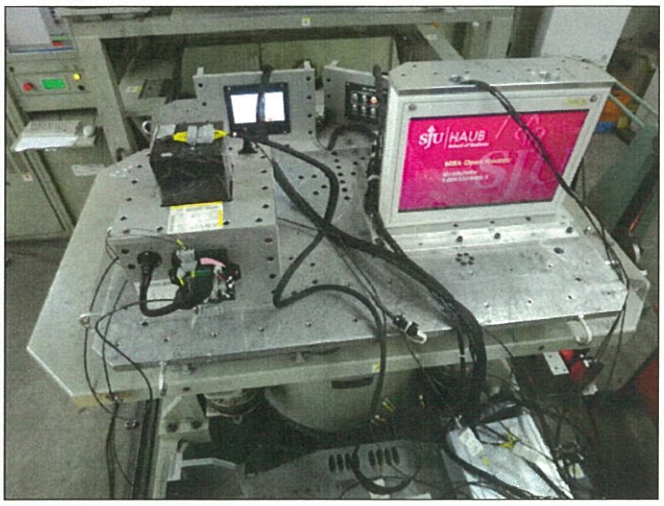
4. Test Scene :



[Longitudinal]

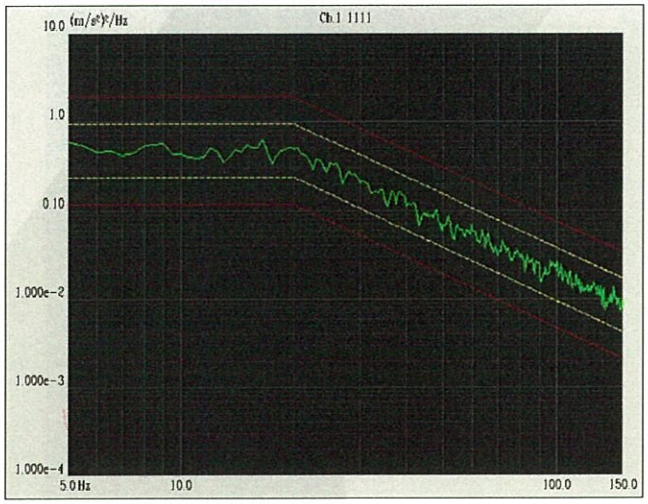


[Transverse]

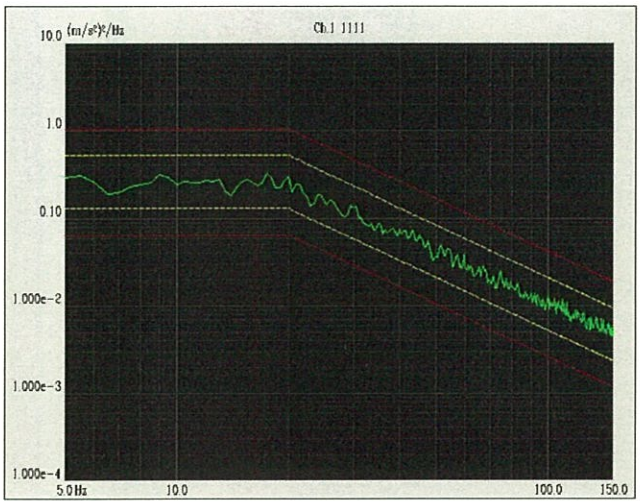


[Vertical]

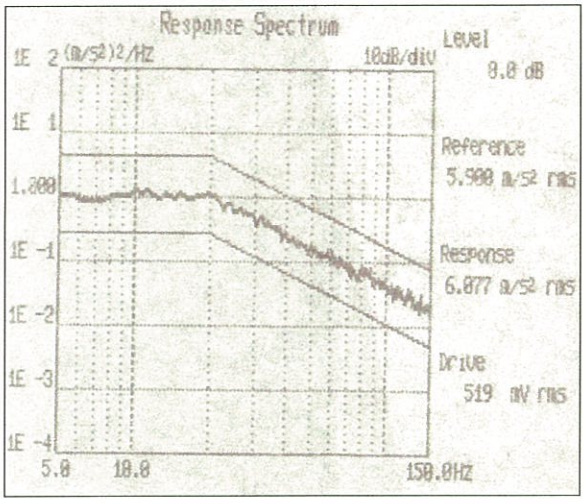
5. Test Graphs



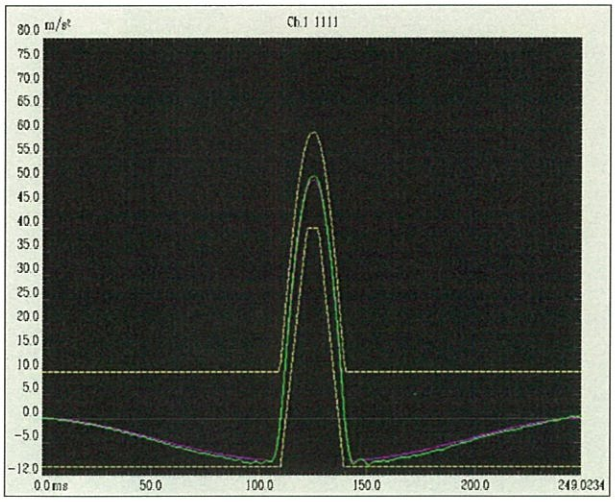
[Graph 1. Longlife Test - Longitudinal]



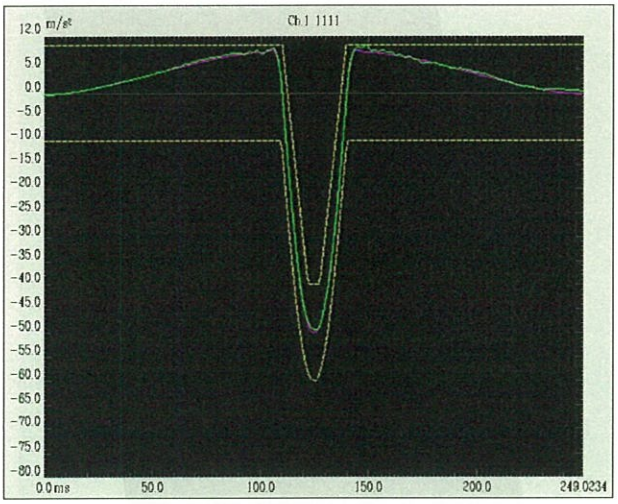
[Graph 2. Longlife Test - Transverse]



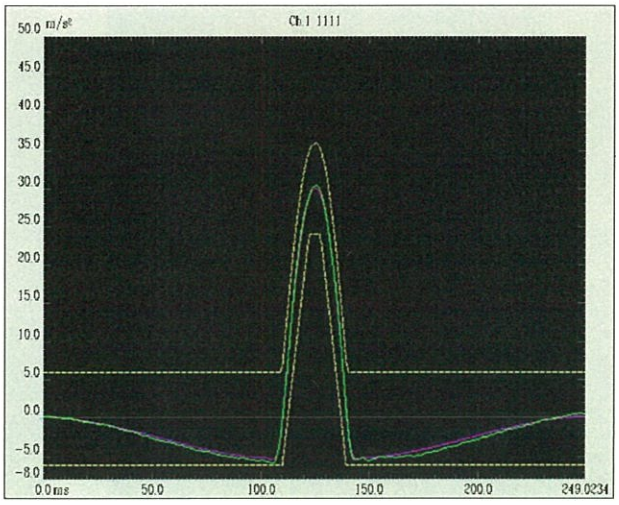
[Graph 3. Longlife Test - Vertical]



[Graph 4. Shock Test - Longitudinal(+)]

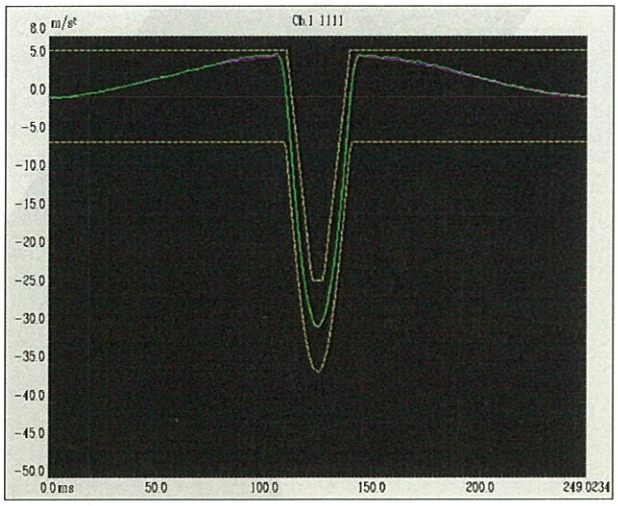


[Graph 5. Shock Test - Longitudinal(-)]

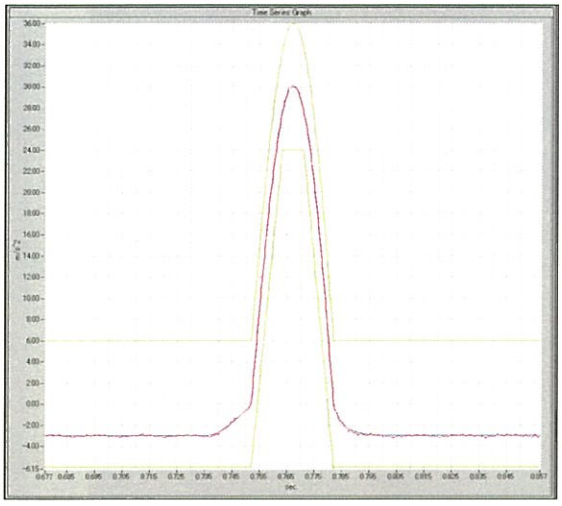


[Graph 6. Shock Test - Transverse(+)]

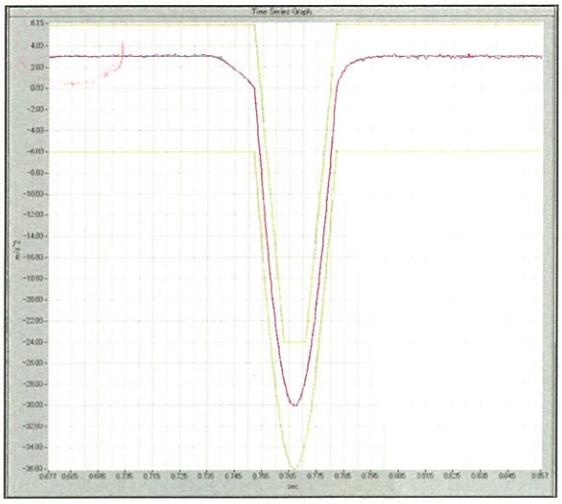
TEST REPORT



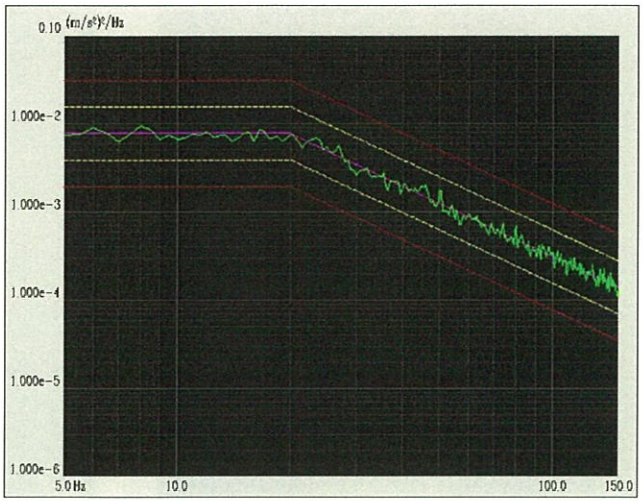
[Graph 7. Shock Test - Transverse(-)]



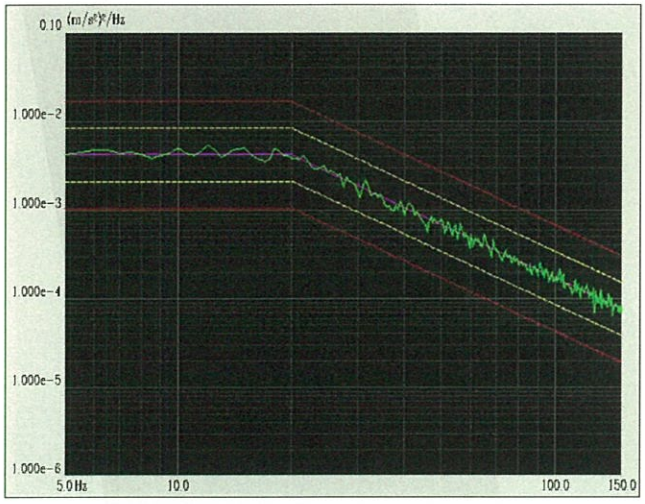
[Graph 8. Shock Test - Vertical(+)]



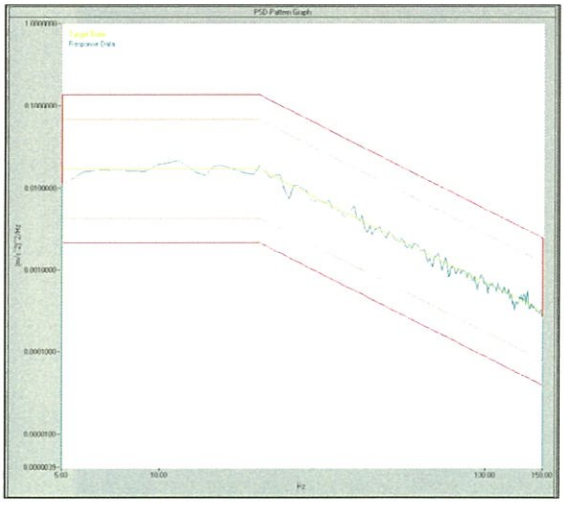
[Graph 9. Shock Test - Vertical(-)]



[Graph 10. Functional Test - Longitudinal]



[Graph 11. Functional Test - Transverse]



[Graph 12. Functional Test - Vertical]

- End of Report -